

What actually changes, and when

AI, robotics, and every job in this company through 2041

This document was commissioned to show how AI and robotics reduce the wage bill. **The research does not support that, and this document says so on the next page rather than burying it.** Crew wages in Baton Rouge are rising faster than this model assumed, the machines that would replace a drainage crew do not exist and are not close, and the jobs genuinely being displaced in 2026 are office jobs - which in this company means estimating, bookkeeping, scheduling, marketing and software.

That is better news than it sounds. It means the hardest thing Scag Scapes owns - crews who can dig a good drain in Baton Rouge clay - is the part technology cannot take, and the part it can take is overhead.

The premise is wrong, and it is worth being wrong in this direction

The request behind this document assumed AI and robotics would push wages down. Every credible source points the other way for this kind of work. Correcting it now matters more than almost anything else in the workbook, because crew labour is the single largest cost line and it runs for fifteen years.

Why trade wages are going UP

- **The local record already beat the model.** East Baton Rouge specialty-trade wages (NAICS 238) went from \$1,320 to \$1,586 a week between 2019 and 2024 - a realised **3.75% a year**. This workbook assumes 3.5%. The '2.45% Louisiana haircut' earlier versions applied has been removed.
- **The government's own automation-aware forecast disagrees.** BLS projects construction labourers **+6%** and equipment operators **+4%** employment through 2035, against **+3%** for all occupations. About 137,000 labourer openings a year.
- **The famous 'robots cut wages' study does not apply.** Acemoglu and Restrepo measured 1990-2007 industrial robots - 39% automotive, 19% electronics. The mechanism is a bolted-down machine repeating one motion. A different yard, soil, slope, fence and buried gas line every job is the opposite of that.
- **Labour supply is contracting, not expanding.** Immigrants were 26.3% of US construction in 2024 and 43% of labourers; enforcement has cut employment among likely-undocumented workers by 1.8 points, and over seven workers now leave employment per detention. Supply contractions raise wages.

Where the displacement is actually happening

- Stanford's study of ADP payroll records - 3.5 to 5 million workers a month - found employment for **22-25 year olds in the most AI-exposed occupations down about 6%**, and young software developers down about **20%**, between 2022 and 2025.
- The adjustment showed up in **employment, not pay**. Firms stopped hiring rather than cutting wages.
- The occupations named: **software developers, customer service representatives, marketing and sales managers**. The least affected were physical, in-person jobs.
- Anthropic's own usage data shows **construction and extraction almost absent** from AI usage, while computer and mathematical work is 30% of usage against 4% of employment.
- **Read that against this company's org chart.** The exposed roles here are estimating, bookkeeping, scheduling, marketing and software. The man in the trench is the safest person on the payroll.

Said plainly, and it belongs in writing: of the four people in this plan, the one whose role AI changes most is Ned's - growth, marketing, software and analysis. Charlie's, Bryan's and Eli's work is comparatively insulated. A blueprint that flattered the person who wrote it would not be worth reading.

What can actually be bought, and what is a video

Every row below was checked against the manufacturer's own material this month. The distinction that matters is not clever-versus-dumb, it is **purchasable-versus-not**.

Technology	What it really is, September 2026	Can Charlie buy it for a Baton Rouge backyard?
Built Robotics autonomous trenching	Retrofits 15-50 tonne excavators. The company has publicly refocused from general construction onto utility-scale solar piling.	No. A mini excavator is 1.5-9 tonnes - wrong machine class by three to ten times, and the wrong market.
Cat Command (remote operation)	Ships on the 313 through 395 class. Expansion announcements cover the 349, 352, 374 and 395.	No. Not offered on the 301.5-310 mini class.
Bobcat RogueX, RogueX2, RX3	Concept machines. No production date announced. Dealer pages that appear to list them carry no price and no inventory.	No. If a salesman implies otherwise, that is the moment to stop listening.
Teleo retrofit teleoperation	Real, and it works - but a human still drives, just from somewhere else. Enterprise-quoted, no list price.	Pointless here. It relocates an operator rather than removing one. With a three-person crew already on site it saves nothing.
Autonomous commercial mowing	Scythe Robotics was acquired in March 2026; Electric Sheep was also acquired. Robot-as-a-service only, economic on large open acreage.	Not this market. Two well-funded players exiting inside a year says the economics were harder than the pitch.
Grade control / depth check	Shipping and dealer-installed. Cat E-Fence/Indicate about \$5,000 on a 6-9 tonne machine. Bobcat Depth Check is a factory option on the E32-E55 class the crews already run , +/- 0.5 inch.	YES - and it is the only earthmoving automation on this page that is real for this work. Buy it as a quality tool.
AI voice agent that books jobs	Shipping and cheap. Jobber's receptionist \$99/mo; Goodcall \$66-208/mo; OnCrew \$49-349/mo.	YES. The easiest yes in the entire plan - see the shopping list.
AI takeoff from a Moasure scan	Does not exist. Every AI takeoff product reads 2D plan sets, not raw site topography.	No - this is a build, not a buy. Which is an opportunity, not a gap.

One warning about the numbers attached to grade control. Trimble's own productivity-study page publishes no percentages, and no independent study exists. Every figure in circulation is vendor-funded. Judge it on Charlie's own callback rate after a season, not on a brochure.

Every job in this company, and what happens to it

This is the centre of the document. Pay direction is the honest expectation for **that role's market rate**, not a promise about any individual.

Role	What it is today	By 2031	By 2036	Pay	The real shift
Crew labourer (where Eli starts)	Digs, hauls, lays pipe and fabric, backfills, repairs sod	The same work, plus reading a depth-check display	The same work. No product exists to replace it	UP 3.5%/yr	Almost none. This is the most protected job in the company - and the reason to say so is that it is the one people wrongly assume is doomed
Equipment operator	Runs the mini-ex and skid steer on feel and experience	Runs it with machine guidance; becomes the person who <i>verifies</i> grade rather than guesses it	Possibly supervises semi-autonomy on the rare open commercial site	UP already +51% over a labourer	From craft to craft-plus-verification. The premium is \$9.41/hr in Baton Rouge and cross-training into it is the cheapest raise the company can give
Crew lead / foreman	Runs the crew, owns the day	Also owns the job-cost feedback loop - what was quoted against what it cost	Carries a wider span because data replaces physical presence	UP	Gains accountability for numbers, not just for the hole
Office admin	Phone, AP/AR, scheduling, paperwork	AI takes the data entry and first-line phone. Person handles exceptions, collections, permits, locates	One person covers 3.5 crews instead of 2	FLAT to up per person - but fewer per crew	Bookkeeper becomes operations coordinator. The role does not vanish; the company just stops needing a second one as early
Estimator	Site visits, measures, quotes	AI-assisted takeoff and a Measure-to-proposal pipeline; more quotes per day	One covers 4.5 crews instead of 3	UP per person , fewer of them	From measuring to closing. The in-home conversation is the part that does not automate, so the job becomes more sales and less arithmetic
COO (Bryan)	Runs everything after a job is sold	Runs the field and the data; safety programme is a legal obligation regardless of technology	Judged on outcomes, not headcount under him	UP with revenue bands	Define the role by thresholds, not by how many people report to him - in a leaner company, span-of-control language goes stale fast
Growth & ops lead (Ned)	Marketing, software, systems, the numbers	The most AI-exposed role here. Must shift from producing output to owning judgment and relationships	Justifies itself on outcomes or it should not exist	Must be earned , not indexed	The honest one. AI does most of what this role does today - so the role has to move up the stack or go
Owner (Charlie)	Sells, estimates, designs, dispatches, answers the phone at 7pm	Sells and designs. Everything else has an owner who is not him	Customer relationships and capital allocation	His own policy decision	The goal of every other row on this page. A business that needs its owner present is worth materially less than one that does not

The wage ladder, and the cheapest raise in the business

+51%

the operating-engineer premium over a construction labourer in Baton Rouge - \$27.99/hr against \$18.58

3.75%

actual East Baton Rouge specialty-trade wage growth per year, 2019-2024 - ahead of the 3.5% this model assumes

~\$9,600

to replace one crew member at \$22/hr, at roughly 21% of salary - before the drag on the two people left in the crew

Occupation (Baton Rouge metro)	Median	75th	90th	What it means here
Landscaping worker	\$17.43	\$17.78	\$21.00	The floor. Scag Scapes does not compete here and should not price like it does
Construction labourer	\$18.58	\$22.72	\$28.45	Charlie's \$20-25 sits around the 60th-85th percentile - a genuine recruiting advantage today
Equipment operator	\$27.99	\$31.00	\$36.88	The single biggest structural lever available. Cross-training a good labourer into the seat is cheaper than hiring an operator and doubles as retention
First-line supervisor	\$37.92	\$45.27	\$50.79	Roughly twice a labourer. This is the crew-lead rung on Eli's ladder

Baton Rouge is nearly twice as construction-intensive as the national average, which means these crews are competing with petrochemical turnaround work for the same people. That sets a hard local floor under wages that has nothing to do with drainage - and nothing to do with robots.

The blueprint, in three horizons

NOW TO 2031 - high confidence

- **Crews do not change.** Same three people, same machines, wages up 3.5%+ a year.
- **The office changes.** AI answers the phone and books; bank feeds and receipt capture end most data entry.
- **Grade control goes on the next machine.** ~\$5,000, judged on callbacks.
- **Lead generation starts shifting** - by 2028 a real share of homeowners are asking an assistant, not a search box. Reviews and the Google profile are the whole defence.
- **Do not buy** autonomous equipment, route optimisation, or a 'GEO agency'.
- The company crosses **\$2M of revenue** - worth far more than anything on this page.

2031 TO 2036 - moderate confidence

- **One admin covers 3.5 crews, one estimator 4.5.** That is the whole office saving, and it is real but modest.
- Semi-autonomous earthmoving *may* appear on large open commercial sites. Residential stays manual.
- The operator's job becomes verification as much as excavation; the premium widens.
- AI-assisted quoting from the company's own Moasure data becomes a genuine differentiator **if it was built early.**
- Expect competitors to have the same office tools by now - which is why half the saving is modelled as competed away.
- The company crosses **\$5M** and the buyer pool changes again.

2036 TO 2041 - nobody knows

- **Anyone giving you a confident answer about 2036-2041 is guessing, including me.**
- Dexterous work in unstructured environments has no solved timeline. Sod repair to a standard a homeowner judges by eye is one of the genuinely hard open problems.
- The insurance question may bind before the engineering one: nobody today will underwrite an autonomous machine cutting a gas service in a backyard.
- The sane hedge is not to predict it. It is to own the customer relationships, the licence, the crews and the equipment - and to be big enough by then that the business is an asset rather than a job.

Notice what actually drives the value in all three columns: crossing \$2M and then \$5M of revenue. That is worth multiples more than every piece of technology on this page combined.

The real risk is not robots. It is how customers find you

45%

of US consumers used an AI assistant for a local business recommendation in the past year - up from 6% the year before

88%

of them still check reviews and profiles before acting. AI is a discovery layer; reputation still closes

\$53

average home-services cost per lead on Google Local Services Ads - it has NOT collapsed, so the \$59 in the model holds to about 2028

What is actually changing

- AI Overviews now appear on about **48%** of tracked queries, up from 31% a year earlier. Where they appear, paid click-through falls sharply.
- But AI assistants still surface only about **1.2%** of local businesses, against **35.9%** visibility through Google's local pack. Coverage is thin and winner-take-all.
- Google is migrating Local Services Ads into the main Google Ads platform from 2026 - bidding and reporting change.
- **Angi is collapsing, not tightening its grip:** revenue down 11%, a \$235M impairment, and active pros down 17% year over year. A marketplace shedding pros cannot extract rent from a licensed local contractor.

What to do about it - all of it cheap

- **Google Business Profile, obsessively complete.** AI assistants read the same structured local data Google does.
- **Review volume and recency.** This is the 88% verification step. It is the single highest-leverage marketing activity for the next five years and it costs nothing but discipline.
- **Schema markup** on the site - LocalBusiness, Service, FAQPage.
- **Specific content, not generic:** 'French drain cost in Baton Rouge clay', 'yard drainage after a Louisiana storm'. Generative answers cite entity-rich, specific pages.
- **Budget: \$0-500 a month, mostly labour.** Do not hire a 'GEO agency' at this stage - the field is too young and nobody has a rigorous method.

The shopping list

	What	Cost	Why, and when
BUY NOW	AI voice agent that books jobs	\$99-208/mo	27% of home-services calls go unanswered and 85% of those callers never ring back. Against a \$59 lead, this breaks even on about two saved leads a month . Nothing else in this plan has that payback.
BUY NOW	Grade control / depth check on the next excavator	~\$5,000 per machine	Bobcat Depth Check is a factory option on the E32-E55 class already in the fleet. Buy it for accuracy and callbacks, not to remove a person.
BUY NOW	Reviews and Google profile discipline	\$0-500/mo , mostly labour	The defence against the lead-generation shift, and it must be built before it is needed.
BUILD	Measure-to-priced-proposal pipeline	Principal's time	No vendor sells this for drainage. Built early it is a genuine advantage; built late it is catch-up.
LATER	Second admin hire	Deferred, not cancelled	AI defers the hire rather than removing the role. Model it as one admin covering 3.5 crews by year 15, not as a person eliminated.
BUDGET FOR	Security review of the in-house software	\$5,000-15,000 once	The moment the app holds customer names, addresses, photos of their property or payment data. Self-built software is an operating-margin play, never a valuation story.
DO NOT BUY	Autonomous or remote-operated earthmoving	-	Does not exist for this machine class or this work. Revisit in 2031, not before.
DO NOT BUY	Route optimisation software	-	The 15-30% drive-time savings are real at eight trucks doing many short stops. At two crews on multi-day jobs in one metro the benefit is near zero.
DO NOT BUY	A 'GEO agency' or an AI-marketing retainer	-	The discipline is a year old and nobody has a rigorous method. Do the free version in-house.

The uncomfortable number: on the workbook's own assumptions the AI stack is a NET COST until roughly year ten, and at its best it is worth about one point of revenue. It is worth doing. It is not worth reorganising the company around.

What the company is worth, after this year's refresh

Two of the three valuation multiples in the workbook have been updated to current transaction data, and one was revised **down**. No AI premium has been added, and the reason is evidence, not caution.

Input	Was	Now	Why
SDE multiple, under \$1M revenue	2.30x	2.49x	BizBuySell rolling closed sales for landscaping and yard service, now 2.49x SDE and 0.72x revenue on a median price of \$449,950. Note pest control - the most contract-recurring home service there is - has converged <i>down</i> to 2.46x: the recurring-revenue premium still does not exist in the earnings multiple.
EBITDA multiple, \$2M-\$5M	4.0x	4.0x	Unchanged. IBBA / M&A; Source Market Pulse confirms 4.0x in this band.
EBITDA multiple, above \$5M	6.0x	5.8x	Revised down. The \$5M-\$50M band reached 5.8x in Q2 2026 - its highest since Q1 2022. The old 6.0x sat above even the observed peak, so it is now set to the peak rather than beyond it.
An AI / systems premium	-	None	Advisors were asked directly: 67% report no material AI impact on valuations , 15% say it is too early, 12% see modest upside, 3% see downside. Any genuine margin improvement already flows through SDE and EBITDA - adding a premium on top would count it twice.

The threshold thesis got stronger, not weaker

87% of deals above \$5M drew three or more offers in Q2 2026 and a third drew ten or more, while the sub-\$2M market still favours buyers. The gap between the bands **widened**. Meanwhile the moat is regulatory and physical and AI does not touch it: Louisiana requires a residential contractor licence above \$50,000 of work, a \$50,000 minimum net worth, \$100,000 of liability cover, workers' compensation and a law exam - and maintains a dedicated **Culverts and Drainage Structures** classification. AI lowers the cost of finding and quoting work. It does not lower the cost of digging a trench in Baton Rouge clay, and it does not issue a licence.

What this means for who owns what

What does not need to change

- **Bryan's 12% stays a straight percentage.** If AI means fewer people produce the same profit, each point of the company is worth *more*, not less. Percentage equity is exactly the right instrument for that.
- **Ned's profits interest is already correct.** It pays on value created above a hurdle - inherently margin-linked and indifferent to headcount.
- **Do not invent a 'profit per employee' vesting metric.** There is no evidence base for it in small contracting, and anyone selling that framework is ahead of the data. What the market actually does when productivity per person rises is give **larger grants to fewer people** - median equity grants at small companies rose about 59% between early 2024 and 2026.

What should change

- **Define the COO role by outcomes and thresholds, not by span of control.** 'Runs four crews and six staff' ages badly in a company where one coordinator covers 3.5 crews. 'Owns field margin, safety and job cost' does not.
- **Put any equity acceleration on EBITDA or enterprise-value thresholds** - crossing \$2M then \$5M with a margin floor - rather than on headcount or team-size milestones.
- **Ned's role has to be re-earned, not indexed.** It is the most AI-exposed seat here. The correct structure is the one already in place: a below-market base and a share of value created, so if the role stops being worth it, it stops costing much.
- **Write the technology decisions into the shared rules**, so buying a \$5,000 grade-control system or a \$99 phone agent is a normal operating decision and not a debate.

One genuine liability to name: an in-house app with no documentation, no tests and one person who understands it is a business-continuity risk on the same order as losing a crew lead - and buyers treat owner-dependence as a reason for earnouts, not a reason to pay more. Boring, hireable technology beats clever technology here.

Where the money actually gets saved

Lever	Roughly worth	How confident	Notes
Crossing \$2M, then \$5M of revenue	Millions, at exit	High - transaction data	The multiple moves from ~2.5x owner's cash to 4.0x then 5.8x EBITDA because the <i>buyer</i> changes. Nothing else on this list is in the same order of magnitude.
Retention of crews	\$50,000-100,000/yr at eight crews	High	Construction churns about 47% a year against 42% for private industry. Replacement runs ~21% of salary, near \$9,600 a head. Halving separations is the second-largest number here and it is entirely within Bryan's control.
Cross-training operators instead of hiring them	\$9.41/hr of premium, captured internally	High - BLS	Turns the largest wage gap in the trade into a retention ladder rather than a recruiting cost.
Tax structure and card strategy	See the Write-offs & taxes tab	Medium - CPA must confirm	Legitimate deductions plus roughly 2% back on fuel and materials routed through business cards, paid in full monthly.
The AI office saving	Negative until ~year 10, then about 1% of revenue	Medium	Real, worth taking, and much smaller than the conversation around it suggests. Half is modelled as competed away, because when every contractor has the same tools the saving becomes lower prices.
Not buying what does not work	Whatever was nearly spent	High	Autonomous equipment, route optimisation at two crews, and AI-marketing retainers. The money not spent on these is a real saving.

The ranking is the point. Two of the top three levers are about people - keeping them and promoting them - in a document about robots.

What nobody knows, stated as such

Everything above is dated and sourced. What follows is not, and it is marked that way deliberately - the value of this document depends on the line between the two being honest.

Genuinely uncertain

- **Whether dexterous outdoor manipulation gets solved at all this century.** Sod repair to a standard a homeowner judges by eye is unsolved, and there is no credible roadmap.
- **Whether cheap sensors matter.** Lidar fell from about \$75,000 to under \$200 a unit in a decade and produced no residential-capable excavator, because the binding constraint is judgment in unstructured settings, not sensor cost. Extrapolating that curve to 'autonomous mini excavators by 2032' is exactly the mistake to avoid.
- **Robot and actuator cost forecasts.** Every one found came from a party selling robots or robot market reports. Treat all of it as marketing.
- **Whether the lead-generation shift arrives as a slope or a cliff.** 45% adoption in a year is fast, but the assistants still surface almost no local businesses. It could stay a slow bleed, or turn quickly.

How to tell if this document is going wrong

- **A dealer quotes a price on an autonomous compact excavator.** That would be the first real signal and it has not happened.
- **Crew wage growth falls below 2.5% a year for two consecutive years** in East Baton Rouge. That would mean the labour thesis here is wrong.
- **Cost per booked job from Google rises above about \$90** sustainably. That is the lead-generation shift arriving early.
- **A competitor in Baton Rouge visibly wins work on price the company cannot match.** That is the commoditisation case, and the response is reputation and speed, not matching the price.
- **Review this page every January.** A blueprint nobody revisits is a document, not a plan.

The single most useful sentence in this document: the thing this business should be most afraid of over fifteen years is not a robot - it is not being able to find and keep people who can dig a good drain, at a price customers will pay.

Sources

Area	Source
Wages & labour	BLS OEWS May 2025, Baton Rouge metro (labourers \$18.58 median, operators \$27.99, supervisors \$37.92) - bls.gov/regions/southwest • QCEW East Baton Rouge NAICS 238, \$1,320 to \$1,586 weekly 2019-2024 - data.bls.gov/cew • BLS Occupational Outlook 2025-35 (labourers +6%, operators +4%) - bls.gov/ooh • JOLTS July 2026 separations - bls.gov/news.release/jolts.t03.htm • Center for American Progress, ~21% of salary to replace
Automation & wages	Acemoglu & Restrepo, 'Robots and Jobs', NBER w23285 / JPE 2020 - nber.org/papers/w23285 • Autor & Thompson, 'Expertise', 2025 • NBER w35129 on immigration enforcement and construction labour supply • NAHB, immigrants 26.3% of construction workforce 2024
Where AI displaces	Stanford Digital Economy Lab, 'Canaries in the Coal Mine', Nov 2025, ADP payroll records - digitaleconomy.stanford.edu • Anthropic Economic Index, June 2026 - anthropic.com/research • METR randomized developer trial (19% slower, believed 20% faster) and its Feb 2026 update - metr.org
Equipment	Built Robotics trenching, 15-50 tonne requirement - builtrobotics.com/solutions/trenching • Cat Command 313-395 class - cat.com • Bobcat RogueX/RX3 concept status - therobotreport.com • Grade control options and pricing - compactequip.com • Trimble Earthworks mini-excavator release, May 2026 - equipmentworld.com • ASI acquires Scythe Robotics, March 2026 - asirobots.com
Rework & operations	Love (2026), ASCE Journal of Construction Engineering and Management - measured rework 0.38% precompletion, 0.76% including postcompletion - asce.org • route-optimisation savings benchmarks - timefold.ai
Lead generation	BrightLocal Local Consumer Review Survey 2026 (45% AI use, 88% verify) - brightlocal.com/research • SearchLight Digital LSA cost per lead, Feb 2026, \$6.72M across 888 contractors - thevalleymarketinggroup.com • Angi Inc. Q2 2026 results - ir.angi.com • Google LSA migration into Google Ads, 2026 - pinmeto.com
Valuation	BizBuySell industry valuation multiples, rolling closed sales - bizbuysell.com/learning-center/industry-valuation-multiples • IBBA / M&A Source Market Pulse Q1 and Q2 2026 (5.8x in the \$5M-\$50M band; 67% of advisors report no AI valuation impact) - prnewswire.com • Carta compensation data - carta.com/data • LSLBC licensing and the Culverts and Drainage Structures classification - lsbc.gov
AI tooling prices	Jobber AI Receptionist, Goodcall, OnCrew, Sameday AI, Smith.ai published pricing, Sept 2026 - pipelineon.com/blog/ai-receptionist-contractor • ServiceTitan 2026 State of AI in the Trades (12% embedded AI) - servicetitan.com/guides/2026-ai-in-the-trades

On confidence. Claims about what exists today and what things cost were verified against primary or manufacturer sources this month. Claims about the next five years are inference from that evidence. Claims beyond ten years are labelled as guesses in the text, because no honest source can do better - and a blueprint that pretended otherwise would be the most expensive kind of document to act on.